Day 4 : Hands-on Subnetting Practical

Task 1 : Given a network address of 192.168.10.0/24, divide it into 4 equal subnets and write down the subnet mask, network address, and broadcast address for each subnet.

Given Network: 192.168.10.0/24

Objective: Divide the given network into four equal subnets and identify the subnet mask, network address, and broadcast address of each subnet.

Step 1: Calculate the Required Subnet Bits

The original network uses a /24 prefix. We need 4 subnets.

Number of borrowed bits:

2^3? No, we need (2^n = 4), so (n = 2).

Therefore, we borrow 2 bits from the host portion.

New prefix: /24 + 2 = /26

Step 2: Calculate the Subnet Mask

The new subnet mask for /26 is:

255.255.255.192

Step 3: Calculate the Block Size

Block size = 256 − 192 = **64**

Therefore, each subnet starts at an interval of 64 in the last octet.

Step 4: Subnet Allocation Table

Subnet	Network Address	Subnet Mask	Broadcast Address
Subnet 1	192.168.10.0/26	255.255.255.192	192.168.10.63
Subnet 2	192.168.10.64/26	255.255.255.192	192.168.10.127
Subnet 3	192.168.10.128/26	255.255.255.192	192.168.10.191
Subnet 4	192.168.10.192/26	255.255.255.192	192.168.10.255

Task 2 : For one of the subnets you created, assign valid IP addresses to 5 devices (such as a router, two laptops, a printer, and a mobile) and list which IP goes to which device.

Selected Subnet: 192.168.10.0/26

Subnet Mask: 255.255.255.192

Usable Host Range: 192.168.10.1 – 192.168.10.62

Broadcast Address: 192.168.10.63

IP Address Allocation Table

Device	Assigned IP Address	Subnet Mask	Default Gateway
Router (LAN Interface)	192.168.10.1	255.255.255.192	N/A
Laptop 1	192.168.10.2	255.255.255.192	192.168.10.1
Laptop 2	192.168.10.3	255.255.255.192	192.168.10.1
Printer	192.168.10.4	255.255.255.192	192.168.10.1
Mobile Phone	192.168.10.5	255.255.255.192	192.168.10.1

Explanation

1. I selected the first subnet, 192.168.10.0/26, from the subnet plan created in Task 1.
2. The usable host range is 192.168.10.1 to 192.168.10.62.
3. I assigned 192.168.10.1 to the router's LAN interface. It acts as the default gateway for the other devices.
4. I assigned unique IP addresses to both laptops, the printer, and the mobile phone.
5. All five devices use the same subnet mask, 255.255.255.192, so they belong to the same subnet.
6. I did not assign the network address (192.168.10.0) or the broadcast address (192.168.10.63) to any device.

Task 3 : Draw a simple diagram (hand-drawn or digital) showing your subnet plan: label each subnet, the range of IPs, and where each device is connected.

Network: 192.168.10.0/24
Subnet Mask: 255.255.255.192

Subnet Plan

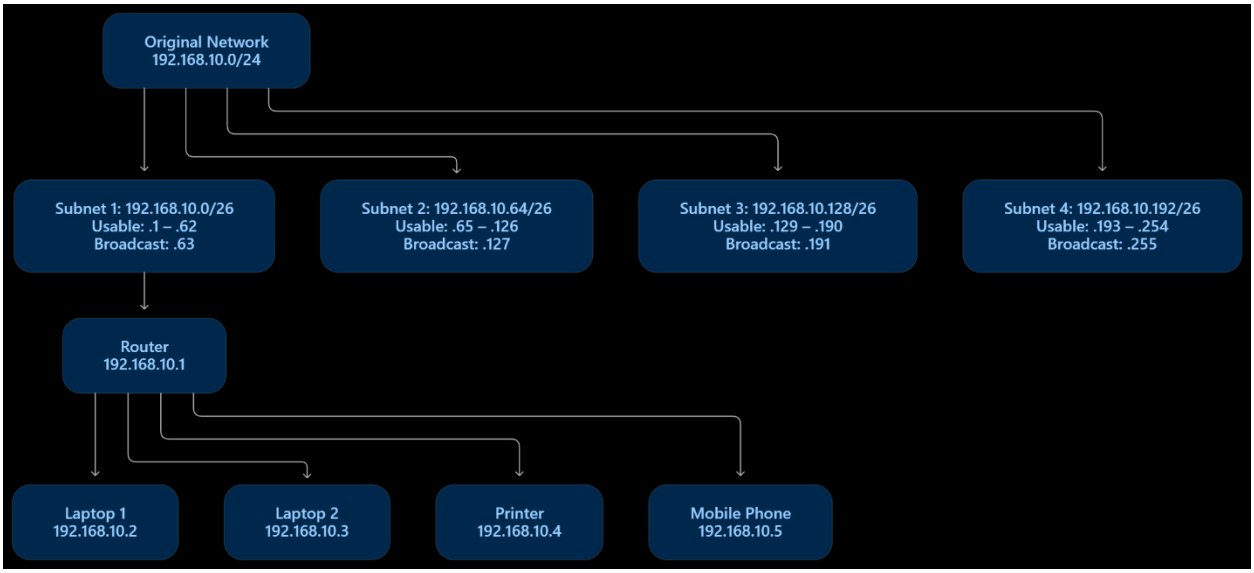
The original network is divided into four equal subnets. Each subnet has 64 total IP addresses and 62 usable host addresses.

- **Subnet 1:** 192.168.10.0/26 — Usable IPs: 192.168.10.1–192.168.10.62
- **Subnet 2:** 192.168.10.64/26 — Usable IPs: 192.168.10.65–192.168.10.126
- **Subnet 3:** 192.168.10.128/26 — Usable IPs: 192.168.10.129–192.168.10.190
- **Subnet 4:** 192.168.10.192/26 — Usable IPs: 192.168.10.193–192.168.10.254

Device Connections

The router's LAN interface uses 192.168.10.1. Laptop 1, Laptop 2, the printer, and the mobile phone are assigned 192.168.10.2, 192.168.10.3, 192.168.10.4, and 192.168.10.5, respectively.

All devices in Subnet 1 use the subnet mask 255.255.255.192. Their default gateway is 192.168.10.1.



Task 4 : Document step-by-step how you calculated the subnet mask, number of hosts per subnet, and assigned IP addresses for your plan, as if you are explaining it to a friend who is new to subnetting.

Step 1: Identify the Original Network

The given network is:

- Network address: 192.168.10.0/24
- Original subnet mask: 255.255.255.0
- Total IP addresses: 256

The /24 prefix means that 24 bits are used for the network portion and 8 bits are available for hosts.

Step 2: Calculate the Number of Subnets Required

We need to divide the original network into four equal subnets.

The formula is:

Number of subnets = 2^n

Here, n is the number of bits borrowed from the host portion.

2^2 = 4 subnets

Therefore, we need to borrow 2 host bits.

Step 3: Calculate the New Subnet Mask

The original prefix is /24. After borrowing 2 bits:

New prefix = 24 + 2 = **/26**

The new subnet mask is **255.255.255.192**.

Step 4: Calculate the Number of Hosts per Subnet

The number of remaining host bits is:

32 − 26 = 6 bits

Total IP addresses per subnet:

2^6 = 64 addresses

Usable host addresses:

64 − 2 = **62 usable hosts**

We subtract two addresses because one is reserved for the network address and one for the broadcast address.

Step 5: Calculate the Block Size

The block size is calculated using the subnet mask's last octet:

Block size = 256 − 192 = **64**

Therefore, the subnet network addresses increase by 64 in the last octet.

Step 6: Identify the Four Subnets

Subnet	Network Address	First Usable IP	Last Usable IP	Broadcast Address
1	192.168.10.0/26	192.168.10.1	192.168.10.62	192.168.10.63
2	192.168.10.64/26	192.168.10.65	192.168.10.126	192.168.10.127
3	192.168.10.128/26	192.168.10.129	192.168.10.190	192.168.10.191
4	192.168.10.192/26	192.168.10.193	192.168.10.254	192.168.10.255

Step 7: Assign IP Addresses to Devices

I selected Subnet 1 for the five devices.

Device	IP Address	Default Gateway
Router LAN Interface	192.168.10.1	N/A
Laptop 1	192.168.10.2	192.168.10.1
Laptop 2	192.168.10.3	192.168.10.1
Printer	192.168.10.4	192.168.10.1
Mobile Phone	192.168.10.5	192.168.10.1

All devices use the subnet mask 255.255.255.192. Each device has a unique IP address within the subnet's usable host range.

Step 8: Verify the Subnet Plan

Before finalizing the plan, I checked that:

1. There are exactly four equal subnets.
2. Each subnet contains 64 total addresses and 62 usable host addresses.
3. No network address or broadcast address is assigned to a device.
4. All five devices have unique IP addresses.
5. The devices use the correct subnet mask and default gateway.
6. The subnet ranges do not overlap.